Physics 1B Formula Sheet

University of California, Los Angeles

Darren Luu

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1 Introduction

This document is a quick reference for all PHYSICS 1B relevant equations, organized both by type and topic. This was made specifically for Dr. Corbin's Winter 2026 PHYSICS 1B lecture.

2 Oscillations

The following contains general forms of various differential equations for oscillations.

2.1 Simple Harmonic Oscillations

The second order differential equation regarding *simple harmonic oscillations (SHO/SHM)* has the following general form:

General Form for Simple Harmonic Oscillations

$$\frac{d^2x}{dt^2} + \omega_0^2 x = 0$$

This equation has the following general solutions:

- $x(t) = A\cos(\omega_0 t + \phi)$
- $x(t) = A\sin(\omega_0 t + \phi)$
- $x(t) = a\cos(\omega_0 t) + b\sin(\omega_0 t)$
- $x(t) = Ae^{i(\omega_0 t + \phi)}$
- $x(t) = Ae^{-i(\omega_0 t + \delta)}$

The most important and useful solution to remember is:

General Solution for Simple Harmonic Oscillations

$$x(t) = A\cos(\omega_0 t + \phi)$$

2.2 Damped Oscillations

The second order differential equation regarding *damped oscillations* has the following general form:

General Form for Damped Oscillations

$$\frac{d^2x}{dt^2} + 2\beta\frac{dx}{dt} + \omega_0^2 x = 0$$

General Solution for Damped Oscillations

$$x(t) = A_0 e^{-\beta t}\cos(\sqrt{\omega_0^2 - \beta^2 t} + \phi)$$

$$x(t) = A(t)\cos(\omega t + \phi)$$

$$A(t) = A_0 e^{-\beta t}, \quad \omega = \sqrt{\omega_0^2 - \beta^2}$$

$$T = \frac{2\pi}{\sqrt{\omega_0^2 - \beta^2}}$$

2.3 Forced Oscillations

The second order differential equation regarding *forced oscillations* has the following general form:

General Form for Forced Oscillations

$$\frac{d^2x}{dt^2} + 2\beta\frac{dx}{dt} + \omega_0^2 x = F\cos(\Omega t)$$

General Solution for Forced Oscillations

$$x(t) = A_0 e^{-\beta t}\cos(\omega t + \phi) + \frac{F}{\sqrt{(\omega_0^2 - \Omega^2)^2 + (2\beta\Omega)^2}}\cos(\Omega t + \xi)$$

Particular Solution:

$$x_{\text{part}}(t) = A(t)cos(\Omega t + \xi)$$

$$A(t) = \frac{F}{\sqrt{(\omega_0^2 - \Omega^2)^2 + (2\beta\Omega)^2}}, \quad \omega = \sqrt{\omega_0^2 - \beta^2}$$

Homogeneous Solution:

$$x_{\text{homog}}(t) = A_0 e^{-\beta t} \cos(\omega t + \phi)$$

Complete General Solution:

$$x(t) = x_{\text{homog}} + x_{\text{part}}$$

2.4 Symbols - Oscillations

- A = Amplitude
  - $A \geq 0$
  - Magnitude of the maximum displacement of the system from equilibrium
- ω_0 = Natural Angular Frequency
  - Rate at which the system passes through phase
  - A constant that converts time to phase
- ϕ = Phase Shift
  - Mathematically shifts the sinusoid in phase (time) to match the physical motion of the oscillator
- $(\omega_0 t + \phi)$ = Phase
  - Angular argument to sinusoid (in radians)
- T = Period (Time for one complete cycle)
  - $T = \frac{2\pi}{\omega_0}$
- f = Frequency (# of cycles transited in unit time)
  - $f = \frac{1}{T} = \frac{\omega_0}{2\pi}$

3 Traveling Waves

The following contains equations for traveling waves.

3.1 General Equation

General Form for Traveling Waves

$$f(x) = A\sin(kx - \omega t + \phi)$$

The above form has the following associated equations:

- $k = \frac{2\pi}{\lambda}$
- $\omega = \frac{2\pi}{T}$
- $\omega = kv_x$
- $f = \frac{1}{T} \rightarrow f\lambda = v_x$

Speed at which disturbances propagate through different media:

- String: $v_x = \sqrt{\frac{T}{\mu}}$;—; T = tension, μ = linear mass density
- Solid: $v_x = \sqrt{\frac{\beta}{\rho}}$;—; β = bulk modulus, ρ = volume mass density
- Gas: $v_x = \sqrt{\frac{\gamma P_0}{\rho_0}}$;—; γ = ratio of specific heats, P_0 = undisturbed, nominal pressure, ρ_0 = volume mass density

3.2 Superposition

Superposition/Sum of Medium Displacements

$$y_R(x,t) = \sum y_i(x,t)$$

Identical Waves in Opposite Directions

$$y_R(x,t) = 2A\sin(kx + \frac{\phi_1 + \phi_2}{2})\cos(\omega t + \frac{\phi_2 - \phi_1}{2})$$

$$*y_R(x,t) = 2A\sin(kx)\cos(\omega t)$$

*Assumes origins in space and time such that the phase-shifts are zero.

3.3 Resonance

Resonant Frequencies

$$f_{N,like} = N\frac{v_x}{2L}$$

$$f_{N,mixed} = (2N+1)\frac{v_x}{4L}$$

3.4 Beats

Wave Equation of Beats

$$y_R(x,t) \approx 2A\cos(\delta\omega t)\sin(\omega_{avg}t)$$

$$\delta\omega = \frac{\omega_1 - \omega_2}{2},\ \omega_{avg} = \frac{\omega_1 + \omega_2}{2}$$

*$2A\cos$: Amplitude envelope, sin: Rapid oscillations

Frequency Equation of Beats

$$f_{beat} = \frac{\delta\omega}{\pi} = |f_1 - f_2|$$

3.5 Doppler Effect

THE Equation

$$\frac{f_{obs}}{f_{src}} = \frac{v_{snd} - v_{obs}}{v_{snd} - v_{src}}$$

4 Electricity

4.1 Coulomb's Law

Coulomb's Law-Electric Force

$$\vec{F}_E = k\frac{q_1 q_2}{r^2}\hat{r} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}\hat{r}$$

4.2 Field Model

Electric Field

$$\vec{E} = k\frac{q}{r^2}\hat{r} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r}$$

$$\vec{F}_E = q\vec{E}$$

4.3 Distributions of Charge

Regarding Various Distributions of Charge

$$\vec{E} = \int d\vec{E} = \int \frac{dq}{4\pi\epsilon_0 r^2}\hat{r}$$

$$dq = \begin{cases} \lambda(\vec{r})ds & \text{for} \quad 1D \\ \sigma(\vec{r})dA & \text{for} \quad 2D \\ \rho(\vec{r})dV & \text{for} \quad 3D \end{cases}$$

4.4 Gauss' Law

REQUIRES SYMMETRY!!!

Gauss' Law

$$\Phi_{\vec{E},cgs} = \oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

There are specific symmetries that should be memorized as shortcuts.

"Gaussian Shortcuts"

Spherical Symmetry: $\rho = \rho(r), \vec{E}(r) = \frac{q_{in}(r)}{4\pi\epsilon_0 r^2}\hat{r}$

Cylindrical Symmetry: $\rho = \rho(r), \vec{E}(r) = \frac{q_{in}(r,x)/x}{2\pi\epsilon_0 r}\hat{r}$

$z \sim\perp$ to plane, $z = 0$ in center-plane: $\sigma(-z) = \sigma(z), \vec{E}(z) = \frac{q_{in}(z,A)/A}{2\epsilon_0}\hat{z}$

4.5 Electric Potential

Potential Difference

$$\Delta V_{AB} = -\int_A^B \vec{E} \cdot d\vec{s} \equiv V_B - V_A$$

$$\Delta U_{AB} = q\Delta V_{AB}$$

Potential Difference in Radial Fields*

$$\Delta V_{AB} = -\int_{r_A}^{r_B} E_r dr$$

*Often shows up in Gauss-friendly distributions

Potential Energy

$$\Delta V_{AB} = -\int_A^B \vec{E} \cdot d\vec{s} \equiv V_B - V_A$$

Potential

$$V(\vec{r}) = \Delta V_{(\vec{r}_{ref},\vec{r})} = -\int_{\vec{r}_{ref}}^{\vec{r}} \vec{E} \cdot d\vec{s}$$

$$\vec{E} = -\vec{\nabla} V$$

4.5.1 Poles, with respect to ∞

Monopole

$$V = \frac{q}{4\pi\epsilon_0 r}$$

$$q = \int \rho d\tau'$$

Dipole

$$V = \frac{p\cos\theta}{4\pi\epsilon_0 r^2}$$

$$p = r' cos\theta' \rho d\tau'$$

Quadrupole

$$V = \frac{Q(\frac{1}{2}(3\cos^2\theta - 1))}{4\pi\epsilon_0 r^3}$$

$$Q = \int (r')^2 \frac{1}{2}(3\cos^2\theta' - 1)\rho d\tau'$$

Octopole

$$V = \frac{O(\frac{1}{2}(5\cos^3\theta - 3\cos\theta))}{4\pi\epsilon_0 r^4}$$

$$O = \int (r')^3 \frac{1}{2}(5\cos^3\theta' - 3\cos\theta')\rho d\tau'$$

4.5.2 Work to Assemble

Collection of Charges

$$W_{ext} = \sum_{i\neq j} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}$$

$W > 0$: Unstable Distribution
$W < 0$: Stable Distribution

Sphere

$$dW_{ext} = dq \cdot V(r)$$

$$W_{ext} = \frac{3Q^2}{20\pi\epsilon_0 R}$$

5 Circuits

5.1 Capacitors

Capacitance

$$C \equiv \left| \frac{Q}{\Delta V} \right|$$

$C = \frac{\epsilon_0 A}{d}$: Parallel Plates
$C = \frac{4\pi\epsilon_0 ab}{b-a}$: Spherical Shells
$C = 4\pi\epsilon_0 a$: Spherical Shell
$C = \frac{2\pi\epsilon_0 L}{\ln(\frac{b}{a})}$: Concentric Cylinders

Series / Parallel

$$\frac{1}{C_{eq}} = \sum \frac{1}{C_i}$$

$$C_{eq} = \sum C_i$$

Dielectrics

$$\vec{E}_{net} = \frac{\vec{E}_{ext}}{\kappa}$$

$$C = \kappa C_0$$

5.2 Energy

Potential Energy

$$U = \frac{1}{2}\frac{q^2}{C} = \frac{1}{2}C\Delta V^2 = \frac{1}{2}q\Delta V$$

Energy Density

$$u = \frac{dU}{dV} = \frac{1}{2}\epsilon_0 E^2$$

5.3 Current

Electric Current

$$I \equiv \int \vec{J} \cdot d\vec{A}$$

$$I = \frac{\Delta V}{R}$$

5.4 Resistors

Resistance

$$R = \frac{\rho L}{A}$$

Series / Parallel

$$R_{eq} = \sum R_i$$

$$\frac{1}{R_{eq}} = \sum \frac{1}{R_i}$$

6 Math

6.1 Coordinate Systems

Cartesian

$$d\vec{s} = dx\hat{x} + dy\hat{y} + dz\hat{z}$$

Cylindrical

$$d\vec{s} = dr\hat{r} + rd\phi\hat{\phi} + dz\hat{z}$$

Polar / Spherical

$$d\vec{s} = dr\hat{r} + r\sin\theta d\phi\hat{\phi} + rd\theta\hat{\theta}$$

6.2 Gradient

Cartesian

$$\frac{d}{d\vec{s}} = \vec{\nabla} = \hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}$$

Cylindrical

$$\frac{d}{d\vec{s}} = \vec{\nabla} = \hat{r}\frac{\partial}{\partial r} + \hat{\phi}\frac{1}{r}\frac{\partial}{\partial \phi} + \hat{z}\frac{\partial}{\partial z}$$

Polar / Spherical

$$\frac{d}{d\vec{s}} = \vec{\nabla} = \hat{r}\frac{\partial}{\partial r} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial}{\partial \phi} + \hat{\theta}\frac{1}{r}\frac{\partial}{\partial \theta}$$